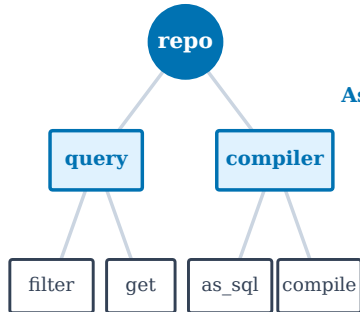


Codebase AST Graph

$$G = (V, E_{\text{call}} \cup E_{\text{caller}})$$

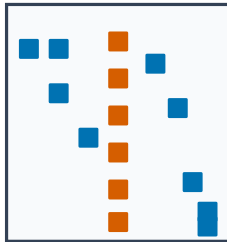


$|V| \approx 10^3 - 10^5$ AST symbols
Power-law: $D_{\text{hub}} \gg D_{\text{fn}}$

Static CSR Matrix

$$W_{uv} = 1.0 \cdot I_{\text{call}} + 0.2 \cdot I_{\text{caller}}$$

Hub column (Logger)



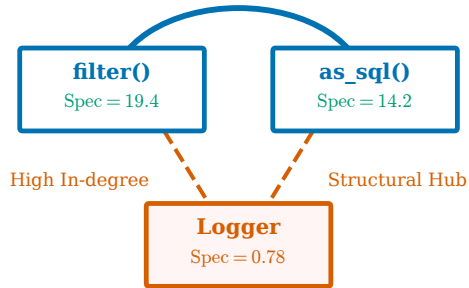
$$T_{uv} = W_{uv} / D_u \quad d_u = I(D_u = 0)$$

Zero-copy mmap binary arrays
< 1.0 ms ingestion overhead

Query-Directed Diffusion

$$\text{Specificity}(v) = \frac{\pi_{\text{query}}(v)}{(\pi_{\text{global}}(v) + \epsilon)^{0.7}}$$

Causal Edge ($\lambda_{\text{call}} = 1.0$)



Suppressed Hub (Spec < 1.0)

$\beta = 0.85$ spectral convergence
95.60% Hub Suppression Index

AST Slicing & Verification

Budget $K_{\text{effective}} = 3,480$ tokens $\leq 4,096$

```
class QuerySet: # query.py:140-158
-   if key in clone.query:
+   parts = key.split('__')
+   if parts[0] in clone.query:

>> ast.parse() OK | pytest: PASSED
```

50% Diagnostic Pass · 0.0% Syntax Errors

Deterministic AST validation (\$0.00 cost)
0.0% syntax errors guaranteed via AST